time flowing, too, as your eyes scan from word to word across the page. Yet, as hard as physicists have tried, no one has found any convincing evidence within the laws of physics that supports this intuitive sense that time flows. In fact, a reframing of some of Einstein's insights from special relativity provides evidence that time does not flow.

To understand this, let's return to the loaf-of-bread depiction of spacetime introduced in Chapter 3. Recall that the slices making up the loaf are the nows of a given observer; each slice represents space at one moment of time from his or her perspective. The union obtained by placing slice next to slice, in the order in which the observer experiences them, fills out a region of spacetime. If we take this perspective to a logical extreme and imagine that each slice depicts *all* of space at a given moment of time according to one observer's viewpoint, and if we include every possible slice, from the ancient past to the distant future, the loaf will encompass all of the universe throughout all time—the whole of spacetime. Every occurrence, regardless of when or where, is represented by some point in the loaf.

This is schematically illustrated in Figure 5.1, but the perspective should make you scratch your head. The "outside" perspective of the figure, in which we're looking at the whole universe, all of space at every moment of time, is a fictitious vantage point, one that none of us will ever have. We are all *within* spacetime. Every experience you or I ever have occurs at some location in space at some moment of time. And since Figure 5.1 is meant to depict all of spacetime, it encompasses the totality of such experiences—yours, mine, and those of everyone and everything. If you could zoom in and closely examine all the comings and goings on planet earth, you'd be able to see Alexander the Great having a lesson with Aristotle, Leonardo da Vinci laying the final brushstroke on the Mona Lisa, and George Washington crossing the Delaware; as you continued scanning the image from left to right, you'd be able to see your grandmother playing as a little girl, your father celebrating his tenth birthday, and your own first day at school; looking yet farther to the right in the image, you could see yourself reading this book, the birth of your great-great-granddaughter, and, a little farther on, her inauguration as President. Given the coarse resolution of Figure 5.1, you can't actually see these moments, but you can see the (schematic) history of the sun and planet earth, from their birth out of a coalescing